

DEPARTMENT OF Computer Science and Engineering



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Minor Project Presentation

On

AI-Powered Solar Tracking and Monitoring Web Platform

Presented By:-

Mudit Sharma 23BCON0599

Jay Solanki 23BCON0574



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Guide:-

Mrs. Surbhi Agarwal

Assistant Professor



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Introduction



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- **Solar energy** is one of the most important renewable energy sources because it is clean, eco-friendly, and easily available. Traditional solar systems use fixed panels, which cannot continuously face maximum sunlight. This reduces energy generation efficiency.
- Our project, **AI-Powered Solar Tracking and Monitoring Web Platform**, solves this problem by automatically tracking sunlight using sensors and servo motors. The system also provides real-time monitoring, weather analysis, and energy prediction through a web dashboard.

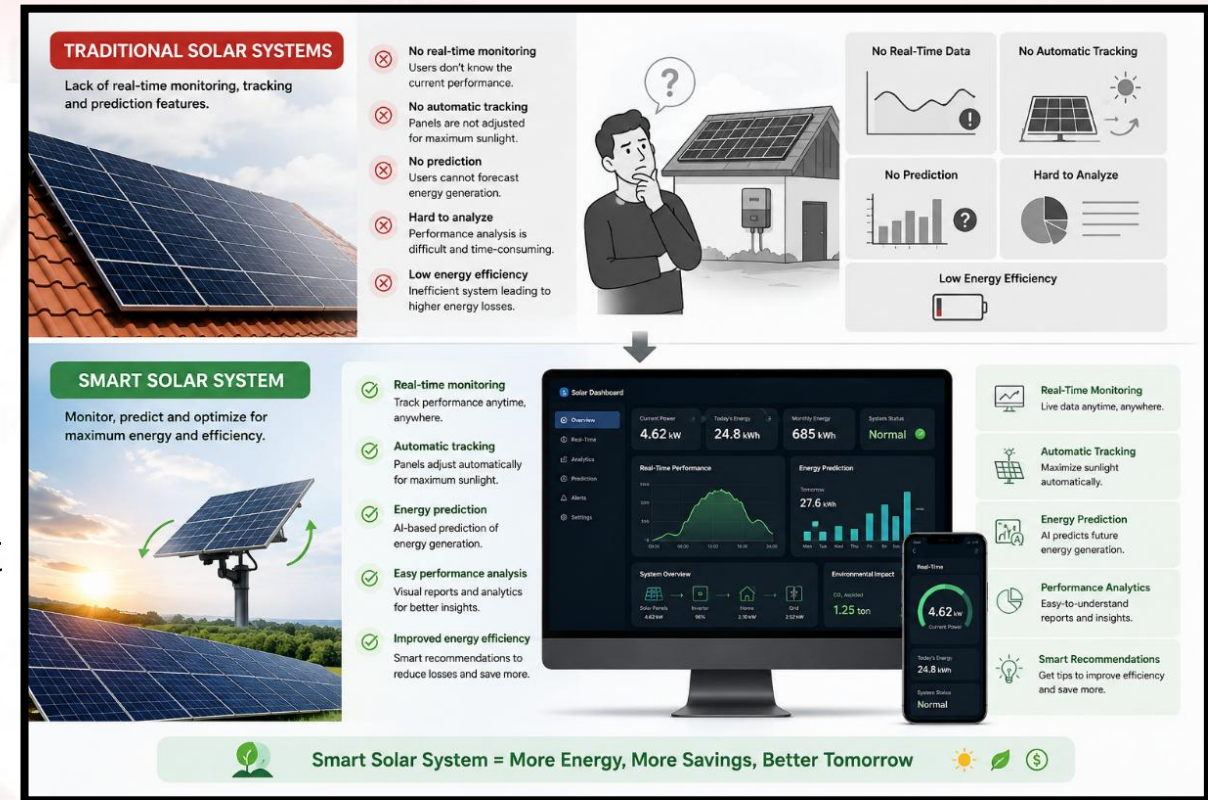


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Problem Statement



- **Traditional solar systems** lack real-time monitoring, automatic tracking, and prediction features. Users cannot easily analyze system performance or improve energy efficiency.
- Therefore, a **smart system** is needed that can monitor solar panels, predict energy generation, and provide intelligent recommendations for better performance.



Literature Review



Several **solar monitoring** systems have been developed using IoT and automation technologies. Existing systems mainly focus on hardware tracking but do not provide AI-based prediction or advanced visualization features.

- The proposed system combines:
- Automatic solar tracking
- Real-time monitoring
- AI-based prediction
- Interactive dashboard visualization



System Design



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The system design of the **AI-Powered Solar Tracking and Monitoring Web Platform** includes hardware, backend, and frontend modules. The hardware layer uses Arduino, LDR sensors, and servo motors for automatic solar tracking. The backend processes sensor and weather data using Python and APIs, while the frontend displays real-time monitoring, graphs, and predictions through a web dashboard.

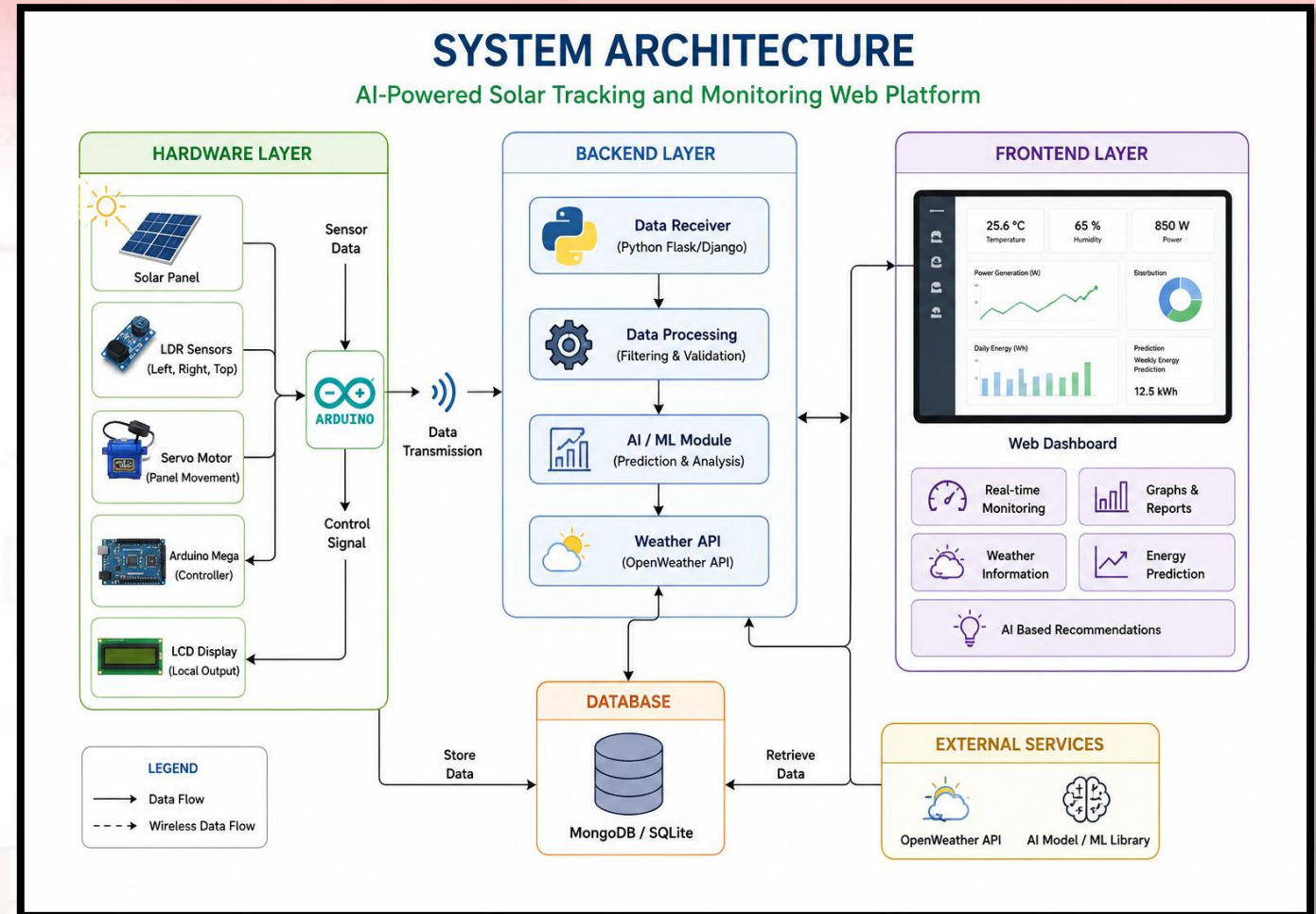
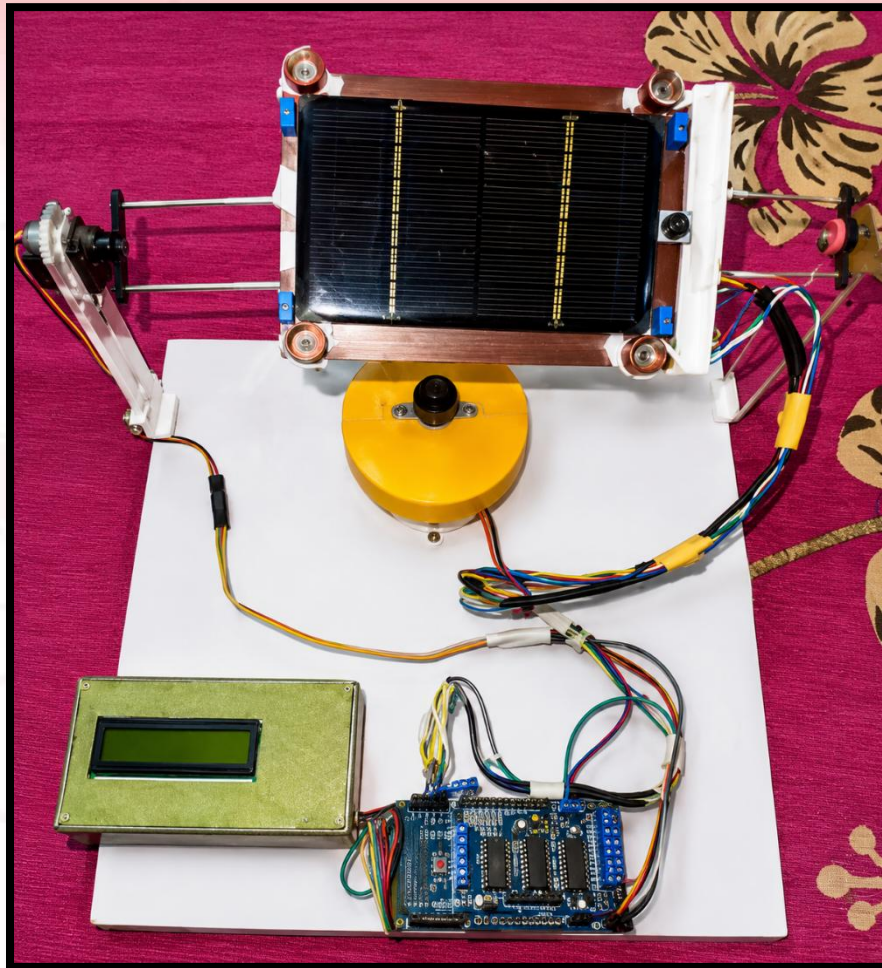


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System Architecture



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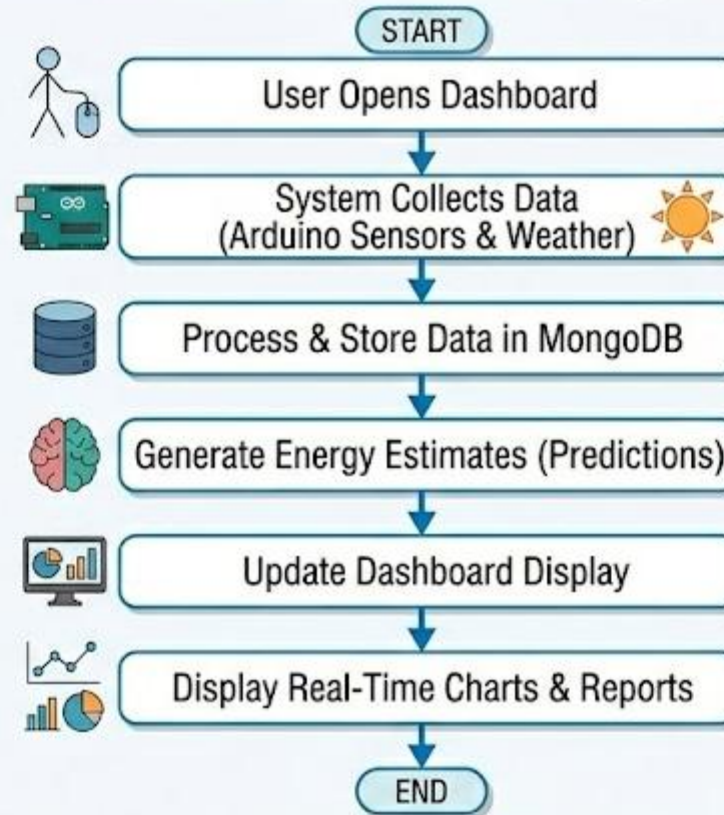


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Flow Chart



Simple Flow Chart: Solar Tracking & Monitoring Platform Data Flow



Methodology



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- The methodology of the project is based on **collecting and processing real-time solar data**. LDR sensors detect sunlight intensity, and the Arduino controller processes sensor values. The servo motor automatically adjusts the solar panel direction toward maximum sunlight.
- Weather data is fetched using APIs and combined with sensor data for prediction analysis. **Machine learning techniques are used to estimate future energy generation**. The processed information is displayed on the web dashboard using graphs and reports.



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Algorithms Used



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1. Light Detection Algorithm

LDR sensors detect sunlight intensity and identify the direction with maximum light.

2. Solar Tracking Algorithm

The servo motor rotates the solar panel toward maximum sunlight for better efficiency.

3. Data Processing Algorithm

Sensor and weather data are processed and stored for monitoring and analysis.

4. Machine Learning Algorithm

Machine learning predicts future energy generation using historical data.

5. Graph Visualization Algorithm

The dashboard displays real-time graphs and energy reports for analysis.



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Tools and Technology Used



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The project uses Python, **Flask**, **HTML**, and JavaScript for developing the web platform and dashboard. SQL/MongoDB is used for database management, while **Kaggle datasets** are used for training and prediction analysis. The **OpenWeather API** is integrated for weather analysis. Libraries such as **Pandas**, **NumPy**, and Scikit-learn are used for data processing and machine learning prediction.

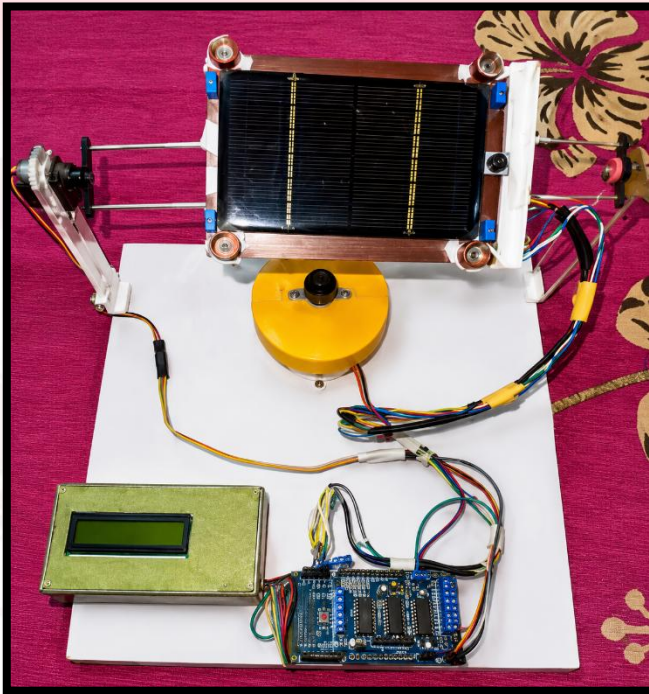


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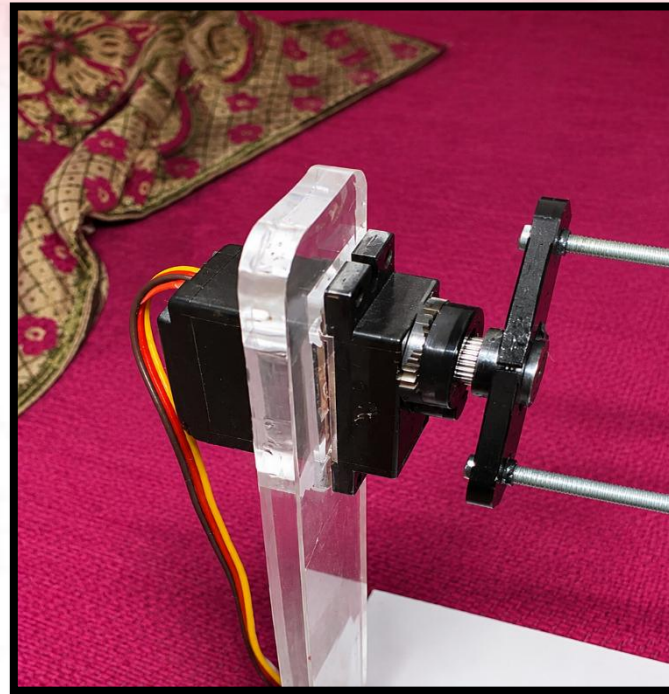
Hardware Components



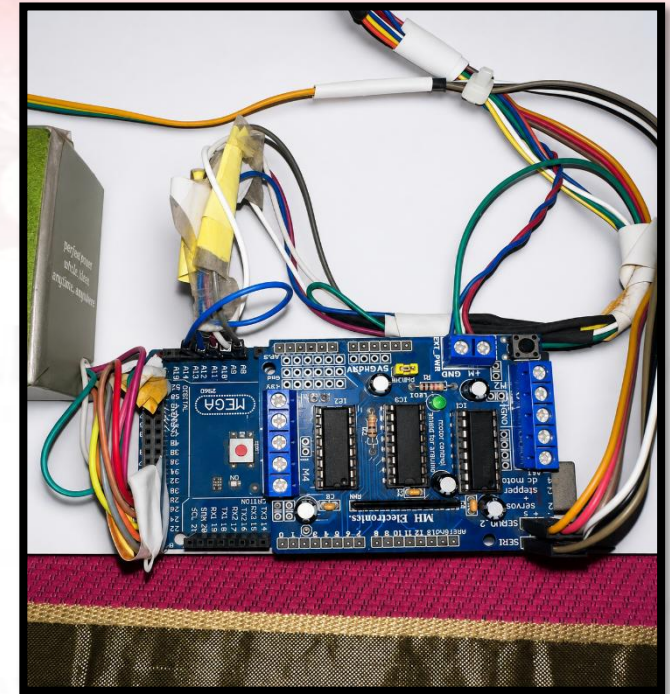
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Solar Panel



Servo Motors



Arduino Mega

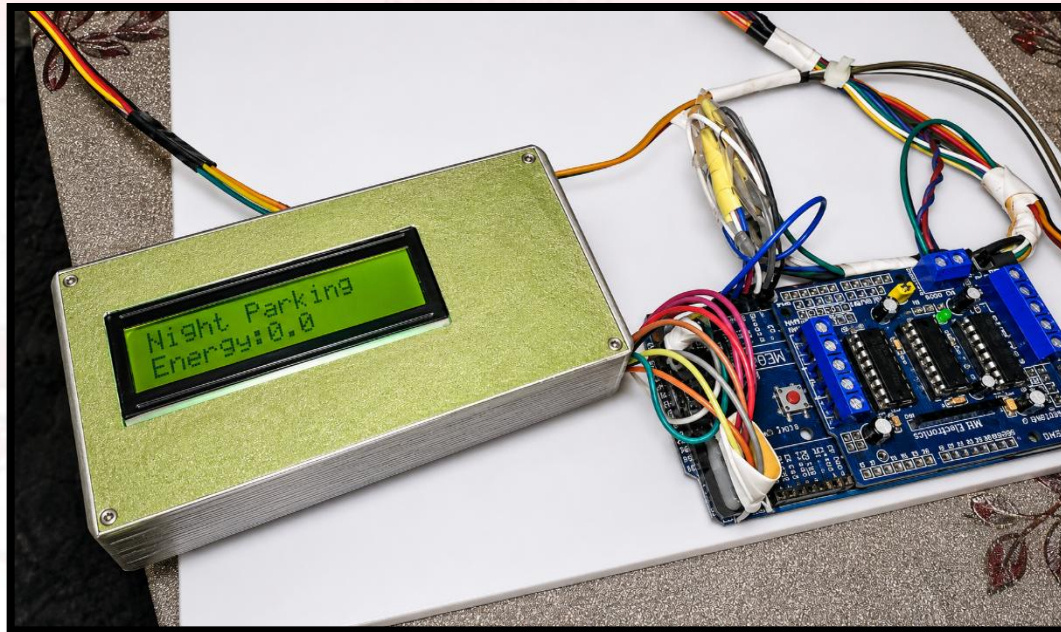


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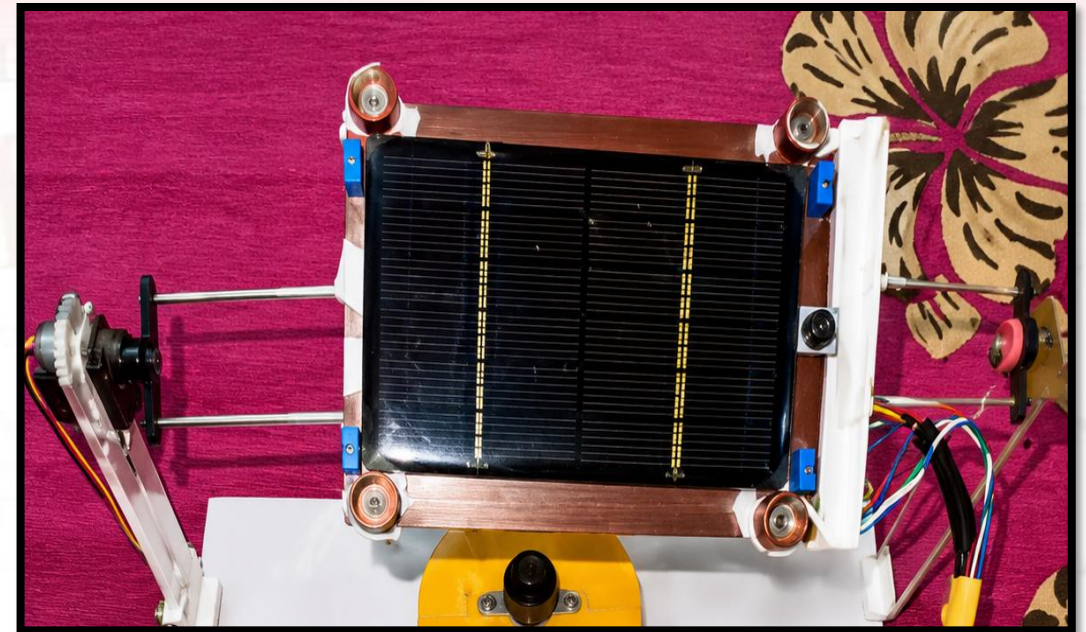
Hardware Components



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LCD Display



LDR Sensors



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Software Components



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- Python
- Flask
- HTML
- CSS
- JavaScript
- SQL / MongoDB
- OpenWeather API
- Kaggle Datasets
- Pandas
- NumPy
- Scikit-learn
- Chart.js
- Visual Studio Code
- Arduino IDE



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Software Components



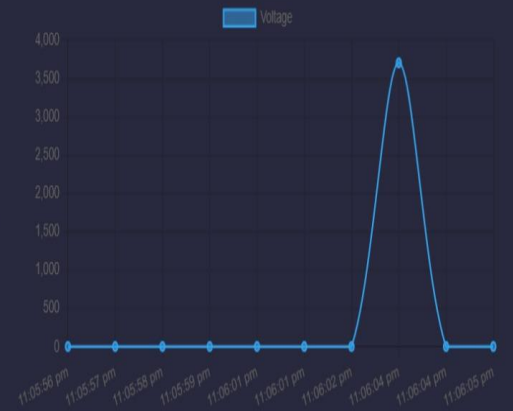
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Real-Time Analytics

Light Intensity Trend



Voltage Output



Real Time Graph

AI Prediction Analytics

Next 24 Hours Prediction



24hr predictive Score



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Roles and Responsibilities



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Mudit Sharma

Worked on hardware integration, Arduino programming, sensor setup, solar tracking mechanism, system testing, and frontend support.

Jay Solanki

Worked on software development, dashboard design, backend integration, database management, and AI prediction module.



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Implementation



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- **Arduino** and **LDR** sensors used for solar tracking
- **Servo motors** used for automatic panel movement
- Python and Flask used for backend processing
- **HTML**, **CSS**, and **JavaScript** used for dashboard development
- **Weather API** integrated for weather analysis
- **Kaggle** datasets used for prediction and analysis
- Real-time monitoring and **graph visualization** implemented

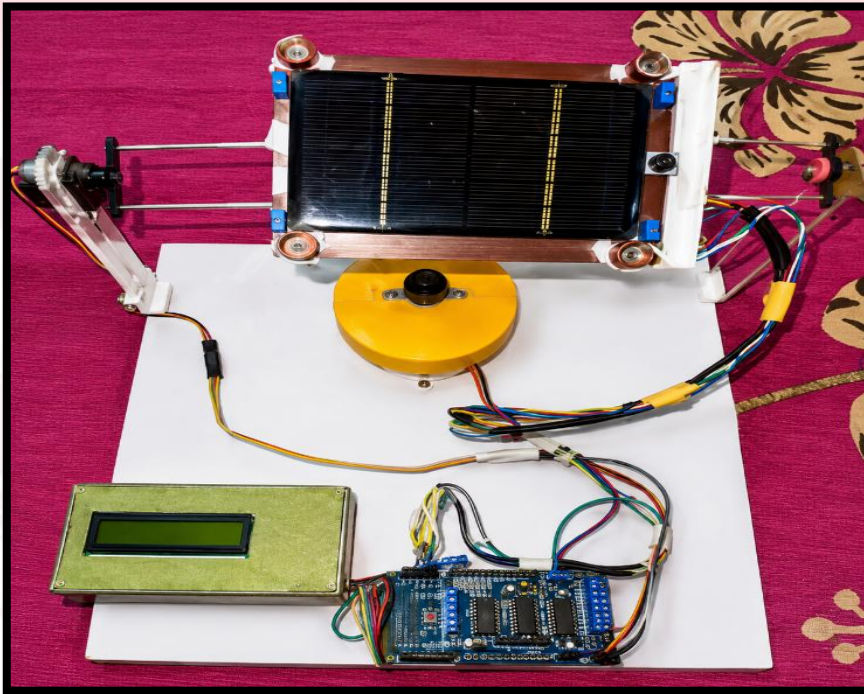


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Results



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Solar Model



Live Dashboard



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Results



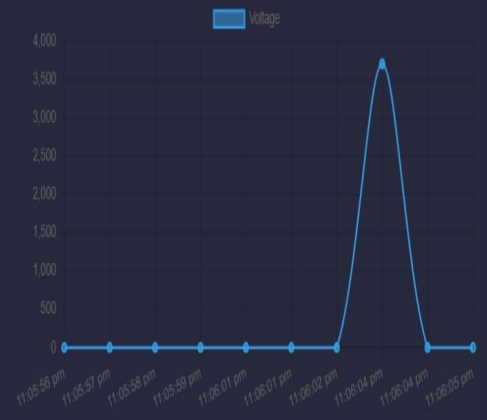
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Real-Time Analytics

Light Intensity Trend



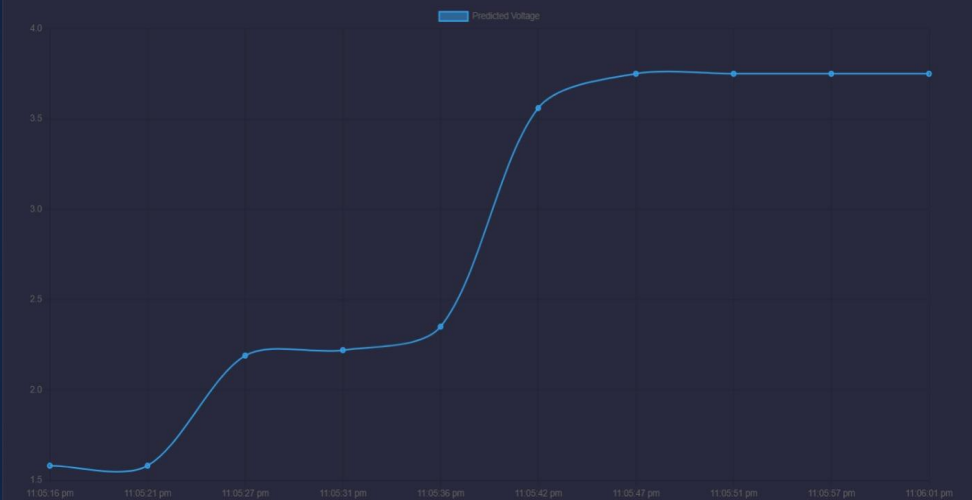
Voltage Output



Real Time Graph

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Next 24 Hours Prediction



24hr predictive Score



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Conclusion



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The project successfully developed a smart solar tracking and monitoring system using IoT, AI, and web technologies. The system provides automatic solar tracking, real-time monitoring, weather analysis, and energy prediction through a web dashboard. It improves solar panel efficiency and helps users analyze system performance more effectively.



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Future Scope



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In future, the project can be enhanced by developing it as a complete smart energy platform named **“SO Bright”**. The system can include mobile application support, advanced AI algorithms, cloud integration, and smart battery management. Features like GPS-based solar tracking, real-time alerts, and smart grid connectivity can also be added to make the platform more intelligent, scalable, and user-friendly.



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References



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- [1] [Python Documentation](#)
- [2] [Arduino Official Website](#)
- [3] [OpenWeather API](#)
- [4] [Kaggle Datasets](#)
- [5] [Flask Documentation](#)



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Thank You



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